Nicola Dell'Avvocato

MISSION ANALYSIS · SYSTEMS ENGINEERING · SPACE

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PROFILE

Space Engineering graduate student oriented towards space missions, space systems and space defence. Ambitious, determined and used to leading multidisciplinary teams towards results. Seeking a career in the international space and defence industry.

EDUCATION

MSc in Space Engineering Sep 2026 – expected Jul 2028
Politecnico di Torino | Turin, Italy

- Orbital Robotics and Distributed Space Systems · Astrodynamics · Space Mission and Systems Design · Space Propulsion · Spacecraft Structures · Spacecraft Dynamics and Control Systems

BSc in Aerospace Engineering Sep 2023 – Jul 2026
Politecnico di Torino | Turin, Italy

- Final Grade: 110/110 with Honours.
- Erasmus+ exchange semester at Universidade da Coruña.
- INTRAPRENDENTI Honour School

Also: Collegio Universitario Renato Einaudi: Merit-based College with Supplementary Training Programme · Scientific High School diploma: Cambridge International Curriculum, 100/100

EXPERIENCE

System Engineer — ICARUS PoliTo (student team) Oct 2026 – present
Politecnico di Torino | Turin, Italy

- System engineering on a solar-powered fixed-wing UAV designed for long-endurance flight on solar energy alone.
- Requirements definition, interfaces between subsystems and verification within a large multidisciplinary team.

Also: Staff of Wave by Vento International Tech Conference · Computer Laboratory Technician · Front Desk Receptionist · Head Lifeguard

TECHNICAL SKILLS

Mission Analysis & Flight Dynamics: Python: RocketPy - Poliastro, Monte Carlo simulation, Simulink, OpenRocket, OpenVSP, ESA DRAMA, Mission Planner

Systems Engineering: Capella MBSE, IBM DOORS

Programming & Numerical Methods: MATLAB: Digital Signal Processing - Aerospace Toolbox, C/C++, JavaScript, HTML, CSS, LaTeX

Simulation & CAD: Abaqus/CAE, SolidWorks, Ansys

Domains of Interest: Space Missions, Satellite Constellations, CubeSats, Earth Observation, UAV Technologies, AI applied to Aerospace

Experimental & Laboratory: Additive Manufacturing, CFRP, Fibre Bragg Grating optical sensors, Dynamic Mechanical Analysis

Transferable: Team Leadership and Coordination, Project Management, Analytical Problem Solving, Critical Thinking, Interpersonal Communication, International Environments

PROJECTS & RESEARCH

Challenge@PoliTO by Leonardo — Simulation & Experimental Validation (upcoming) Feb 2027 – Jun 2027
Politecnico di Torino & Leonardo | Turin, Italy

- Analysis and optimisation of electromagnetic and thermal signatures in complex physical systems.
- Development and validation of numerical models and experimental testing techniques, combining simulation, laboratory testing and data analysis in a real-world research environment.

F450 Quadcopter (personal project, in progress) Oct 2026 – Dec 2026
Personal project | additive manufacturing, autonomy and navigation

- Design and build of an F450-class quadcopter with additively manufactured structural parts, integrating airframe, avionics and software.

- Development towards autonomous flight, including indoor operation where satellite navigation is unavailable.
- Integration of an AI chat interface that translates natural-language input into autonomous mission configurations.

Embedded FBG Sensors for SHM of CFRP Sandwich Structures (BSc thesis) Jun 2026
Politecnico di Torino | Turin, Italy

- Built an Abaqus/CAE finite element model of a composite panel to predict its natural frequencies and strain mode shapes, with an optical fibre embedded between the carbon plies.
- Ran an impact-hammer test campaign with an optical interrogator and processed the strain signals through FFT in MATLAB, comparing measurements against the model.

Also: EUSPA Space for Peace and Resilience with Copernicus/Galileo Satellites Data · Dynamic Characterisation of 3D-printed Metamaterials for Aeroelastic Applications

LANGUAGES, CERTIFICATIONS & HONOURS

Languages: Italian (native) · English, professional working proficiency · Spanish, limited working proficiency

Certifications: B2 First (English), Cambridge International Education · Cambridge International Curriculum IGCSE (A^* Chemistry, A^* Latin) · B2 DELE (Spanish), Universidade da Coruña · BLSD/AED, Croce Rossa Italiana · Lifeguard Certification, Federazione Italiana Nuoto